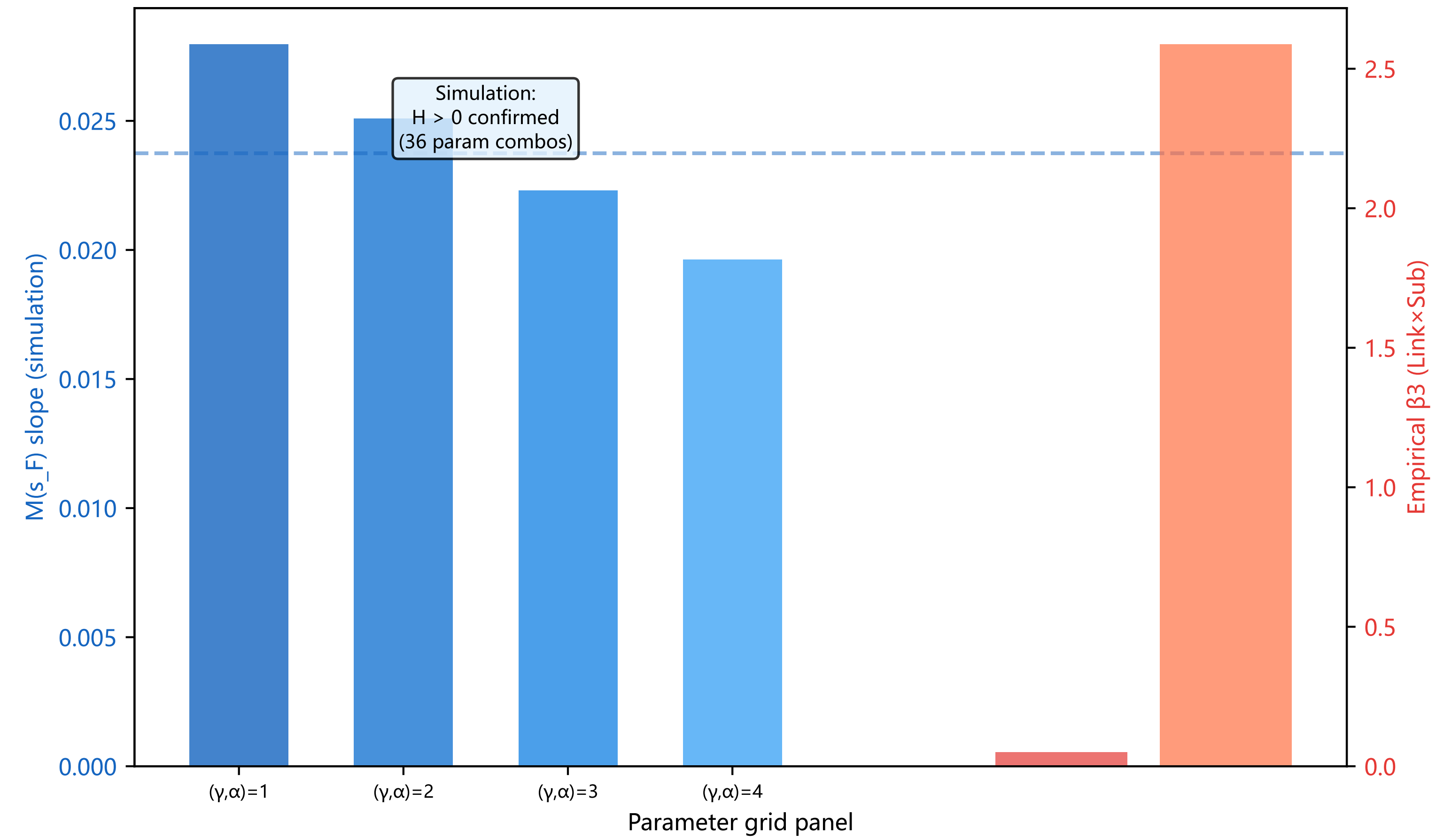
Theory-vs-Empirics Qualitative Comparison: NEV Spatial Agglomeration

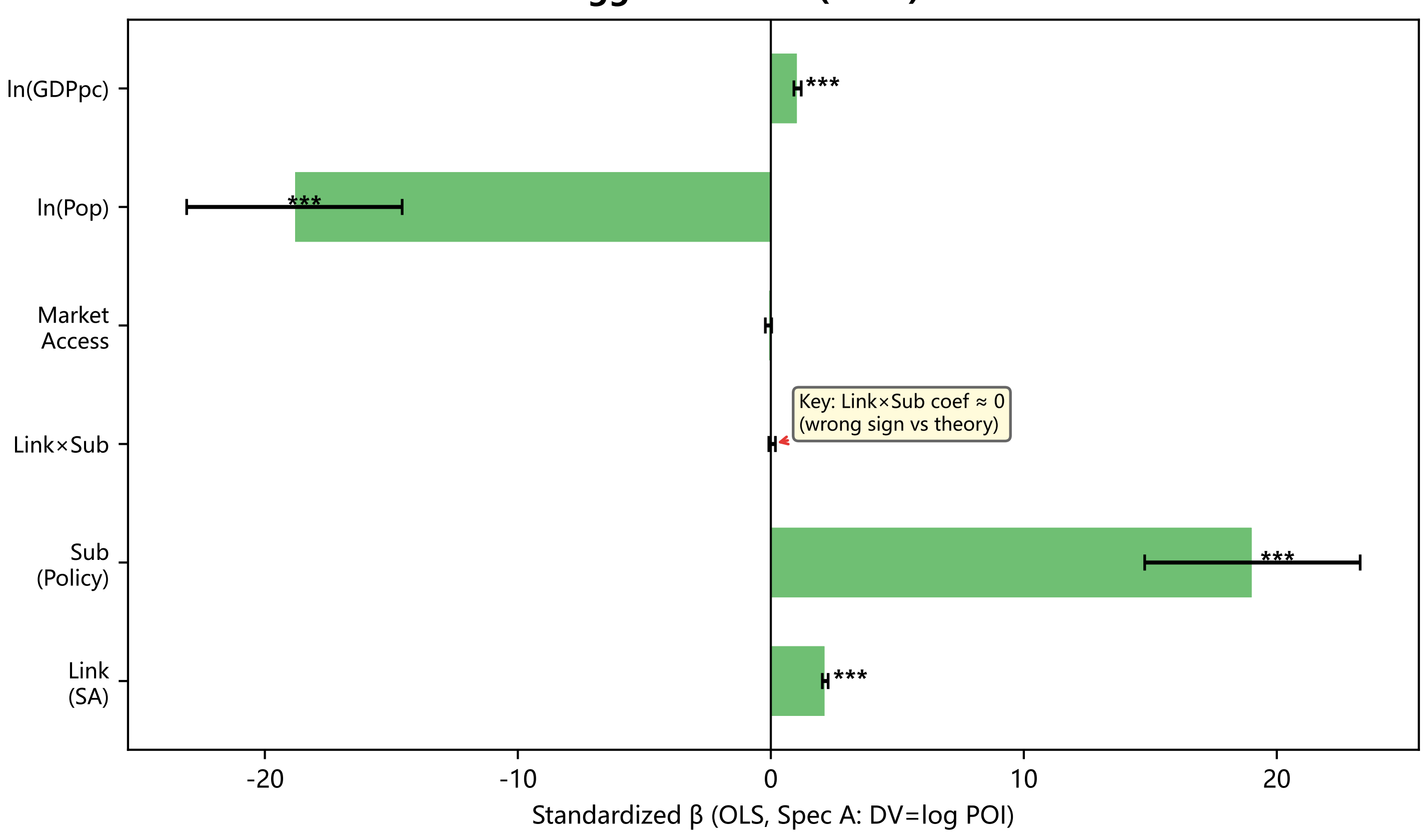
A: Hypothesis Test Summary

Hypothesis	Theory Prediction	Empirical Result	Consistency
H1: Forward Linkage	b1 > 0 (+)	b1 > 0 (+) ***	[OK] Consistent
H2: Subsidy Main Effect	b2 > 0 (+)	b2 > 0 (+) ***	[OK] Consistent
H3: Link x Sub Interaction	b3 < 0 (-)	b3 ~ 0 or > 0 (+) ***	[!!] Contradicted
H4: Spatial Heterogeneity	East: H1 dom. West: H2 dom.	Not yet tested	-- Pending
H5: Spatial Competition	theta_Sub < 0 (-)	SDM rho unstable	-- Pending

B: Cross-Partial H (Simulation) vs β3 (Empirical)



C: OLS Coefficients — NEV Agglomeration (2023)



D: Mechanism Summary

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THEORY-EMPIRICS MECHANISM MAPPING
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Structural Model (NEG)              Empirical SDM
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λ_M (M-firm share)        ←——→    log(1+NEV_POI)
s_r (subsidy rate)        ←——→    GWR policy mentions
P_{B,r} (battery price)   ←——→    1 / Supplier Access
n_{B,r} (B-firm count)    ←——→    Supplier Access (SA)
τ_B (trade cost)          ←——→    1/W (spatial weights)

KEY FINDINGS:
[OK] H1 Forward Linkage: Strongly confirmed (p<0.001)
[OK] H2 Subsidy Main: Confirmed for firm agglomeration
[!!] H3 Substitution: EMPIRICALLY REVERSED
   Model: d2Y/(dLink*dSub) < 0 (substitutes)
   Data:  d2Y/(dLink*dSub) >= 0 (complements)

-> Implication: In China's NEV industry,
  subsidies and supply chains are COMPLEMENTS
  not substitutes -- governments support cities
  that already have industrial foundations.
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